Kazakh-British Technical University

Data collection and preparation

STUDENTS' INDEPENDENT STUDY (SIS) №1

Climate and Population Growth of Kazakhstan Cities

Full Name: Nurzhan Kalmurat

ID: 24B031138

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Almaty

1. Goal of the project

Kazakhstan is a very big country: the south is warm and the north is very cold in winter. Also some cities grow very fast and some cities lose people. In this project I collected real data from two different sources, cleaned and merged it, and checked three questions: **(1) how the climate changes from the south to the north, (2) which cities have the most extreme climate, (3) if the climate of a city is connected with its population growth from 1999 to 2023.**

2. Data collection

Source	How I collected it	What I got
Wikipedia "List of cities in Kazakhstan" (static web page)	requests + BeautifulSoup: checked robots.txt, sent User-Agent, parsed <table class="wikitable"> row by row	32 cities: population 1979-2023, foundation year, status, province
Open-Meteo Geocoding API (REST, JSON)	GET request for every city, filter country = KZ and populated place, alias for 4 names	latitude, longitude, elevation
Open-Meteo Historical Weather API (REST, JSON)	GET with query parameters, timeout, retry with backoff for 429/5xx, pause between requests	11 680 daily rows (32 cities x 365 days of 2025), 7 weather variables

3. Cleaning and merging

- **Scraped table:** removed footnotes and not needed columns, renamed columns to snake_case, removed commas and changed text to numbers (Int64), took only the year from "City since" ("ancient", "Middle Ages" -> NaN), made 19 different province names the same (old name South Kazakhstan -> Turkistan) and added a macro region (South, West, Central, East, North).
- **Missing values:** pop_1979 is empty for 3 cities, so I used 1999 -> 2023 for growth (no missing values). The weather had no missing values and no duplicates; I still added interpolation for small gaps.
- **Quality checks:** 365 days for every city, min <= mean <= max for all rows, no negative rain. IQR outliers are only in rain, snow and wind - these are real weather events, so I kept them.
- **New features:** growth % and yearly growth rate (CAGR); for every city: year, January and July temperature, July - January range, frost days (min < 0 °C), hot days (max > 30 °C), year rain and sunshine.
- **Merging by city:** Wikipedia + geocoding + climate with validate="one_to_one" and indicator=True (32 of 32 matched), daily weather + city table with validate="many_to_one".

4. Results

Climate from south to north. Latitude is the main factor: the correlation between latitude and year temperature is r = -0.90. A linear regression shows that every 1° to the north the year is about 0.9 °C colder ($R^2 = 0.82$). In 2025 the South had 13.5 °C on average and the North only 5.8 °C. In January the South is about -3 °C and the North and Central regions about -10 °C. The North has about 170 frost days and 9 hot days, the South 114 frost days and 99 hot days. Elevation has almost no link with temperature, because the high cities (Almaty, Taraz) are also in the south.

Extreme cities. Turkistan has the hottest summer (July mean 31.6 °C, max 43.1 °C). Schuchinsk, Astana and Saran have the coldest winter (January about -10.5 °C, Saran had -33.5 °C). Balqash has the biggest difference between July and January (36.1 °C), and Aktau on the Caspian Sea is the mildest city (only 65 frost days).

Climate and population. Cities with a warmer winter grew faster: Spearman rho = 0.63 ($p < 0.001$) between January temperature and growth 1999-2023. The median growth was +121% in the West and +77% in the South, but -5.5% in the Central region. All 7 cities that lost people are cold northern, central or eastern towns, most of them old mining towns (Arkalyk -37%, Saran -29%, Shakhtinsk -29%).

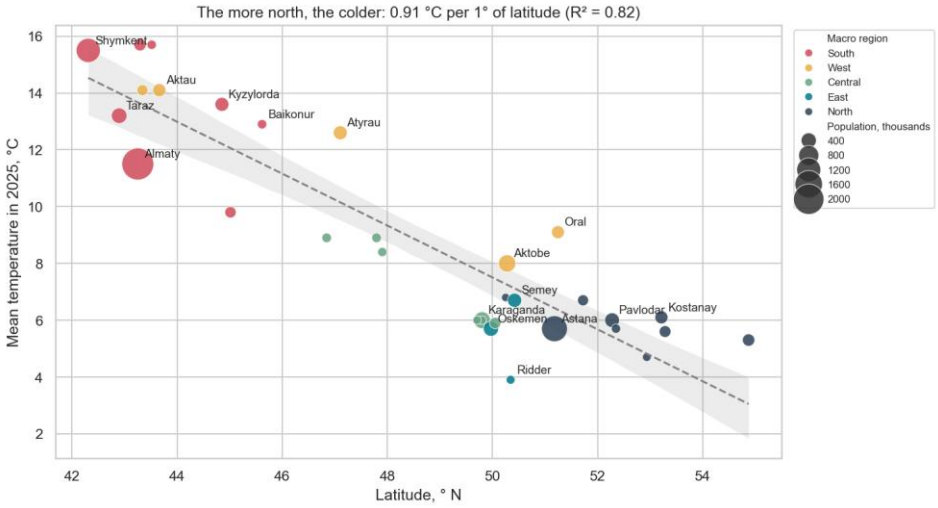


Figure 1. Latitude vs mean year temperature of 32 cities (2025)

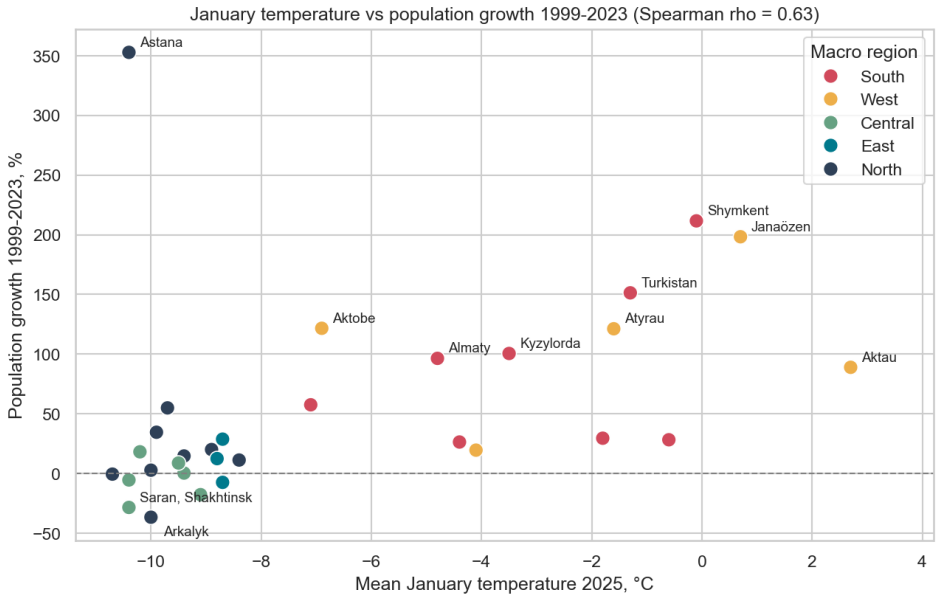


Figure 2. Mean January temperature vs population growth 1999-2023

5. Key insights and conclusion

- **Geography decides the climate:** the further north, the colder and the longer the winter. The sea (Aktau, Atyrau) makes winter softer, mountains (Ridder) make the city colder.
- **Population moves to the warm south and west,** and cold industrial towns lose people.
- **Correlation is not causation:** Astana is one of the coldest cities but grew +353% because it is the capital; western cities grow because of oil, the South has a high birth rate. Climate is only one factor together with economy and politics.

Limitations: weather for only one year (2025), model data on a grid (not weather stations), 32 cities from one Wikipedia table, one point for the whole city. Next step: use 10-30 years of weather and add economy data (World Bank, stat.gov.kz) for a regression with several factors.